CDE4301 IS-431

Redesigning the Hotel Arrival Experience

Interim Presentation



Description: Citadines Connect City Centre Singapore
Image Credit: The Ascott Limited



Description: Opening Scene
Image Credit: Generated by IS-431



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Problem Statement

Invisible Hospitality:

How can we reimagine the hotel arrival experience with AI and Emerging Technologies?

01

Preliminary Research



Description: Citadines Science Park Singapore
Image Credit: The Ascott Limited

Industry Overview

Luxury	Ascott	Oakwood Premier	Crest Collection	
	Raffles	Mandarin Oriental	Shangri-La	The Ritz-Carlton
Mid-range	Citadines	Somerset	Oakwood	Unlimited Collection
	Novotel	Holiday Inn	Hilton Garden Inn	Furama
Budget	lyf			
	Hotel 81	ibis budget	ibis	

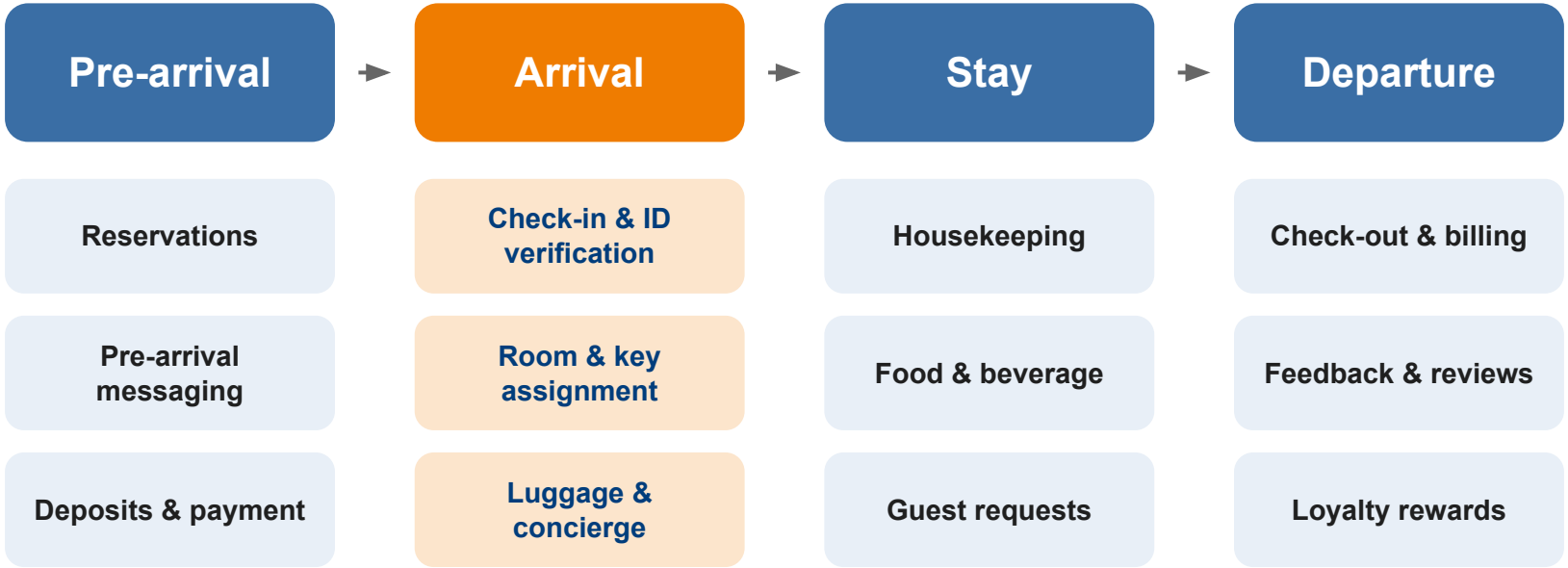
Ascott group

Industry Overview

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Ascott group

Core Services Hotels Provide



Manpower Shortage in Singapore Hotels

4th

Hardest-to-fill job

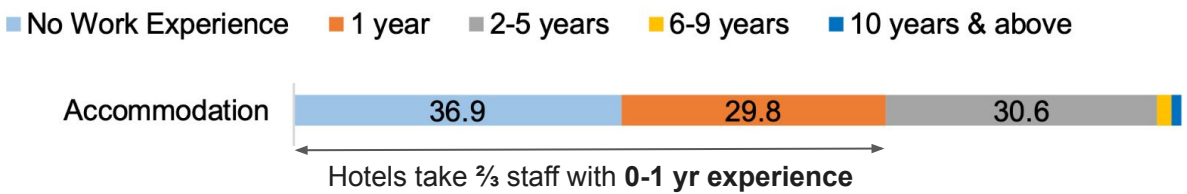
< 35%

Cap on foreign workers
out of total workforce

S$3,600

Min salary
for new S Passes
from 1 January 2027

Chart 9: Distribution of Job Vacancies by Minimum Work Experience Required and Industry, 2025 (%)



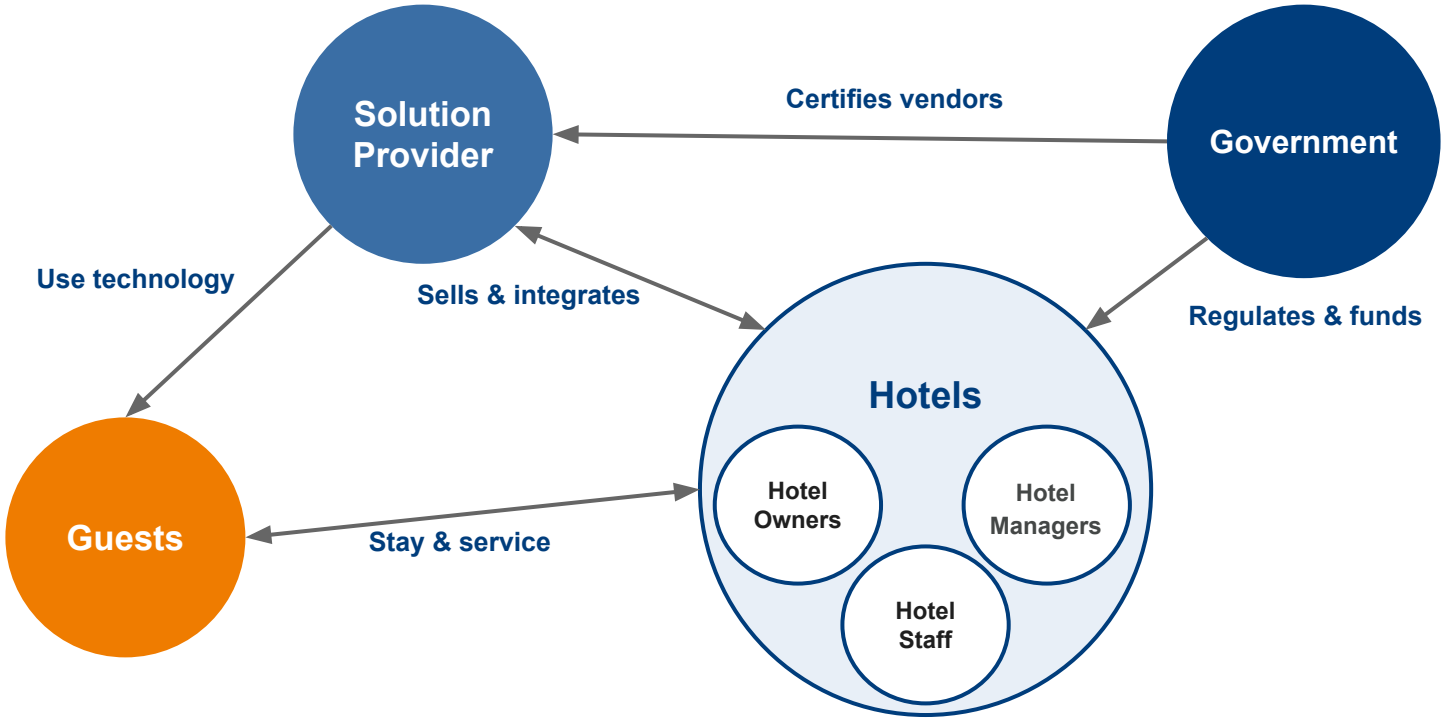
Source: Job Vacancy Survey, Manpower Research & Statistics Department, MOM
Career Jobs Singapore Hospitality Food & Beverage Industry, Andaz Singapore
Foreign Workforce Policy Announcements, MOM

Industry Overview

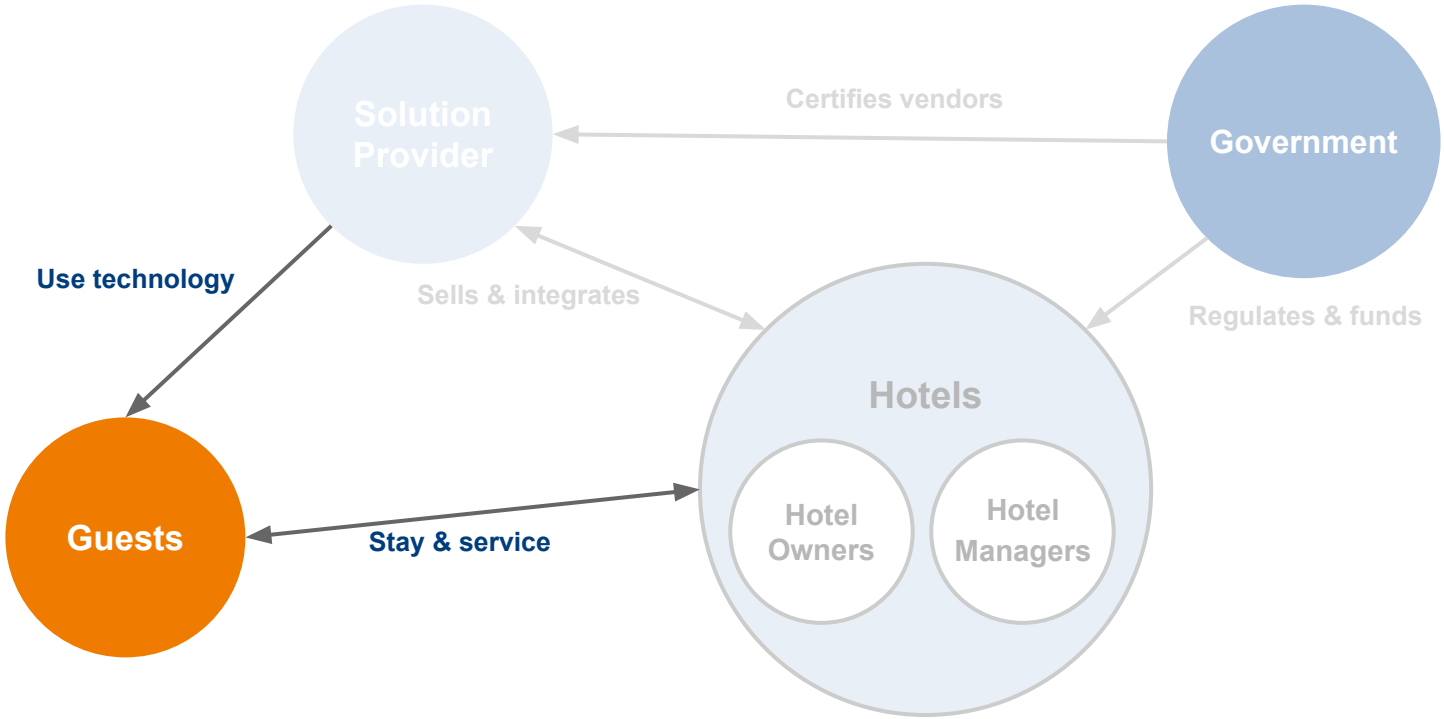
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Ascott group

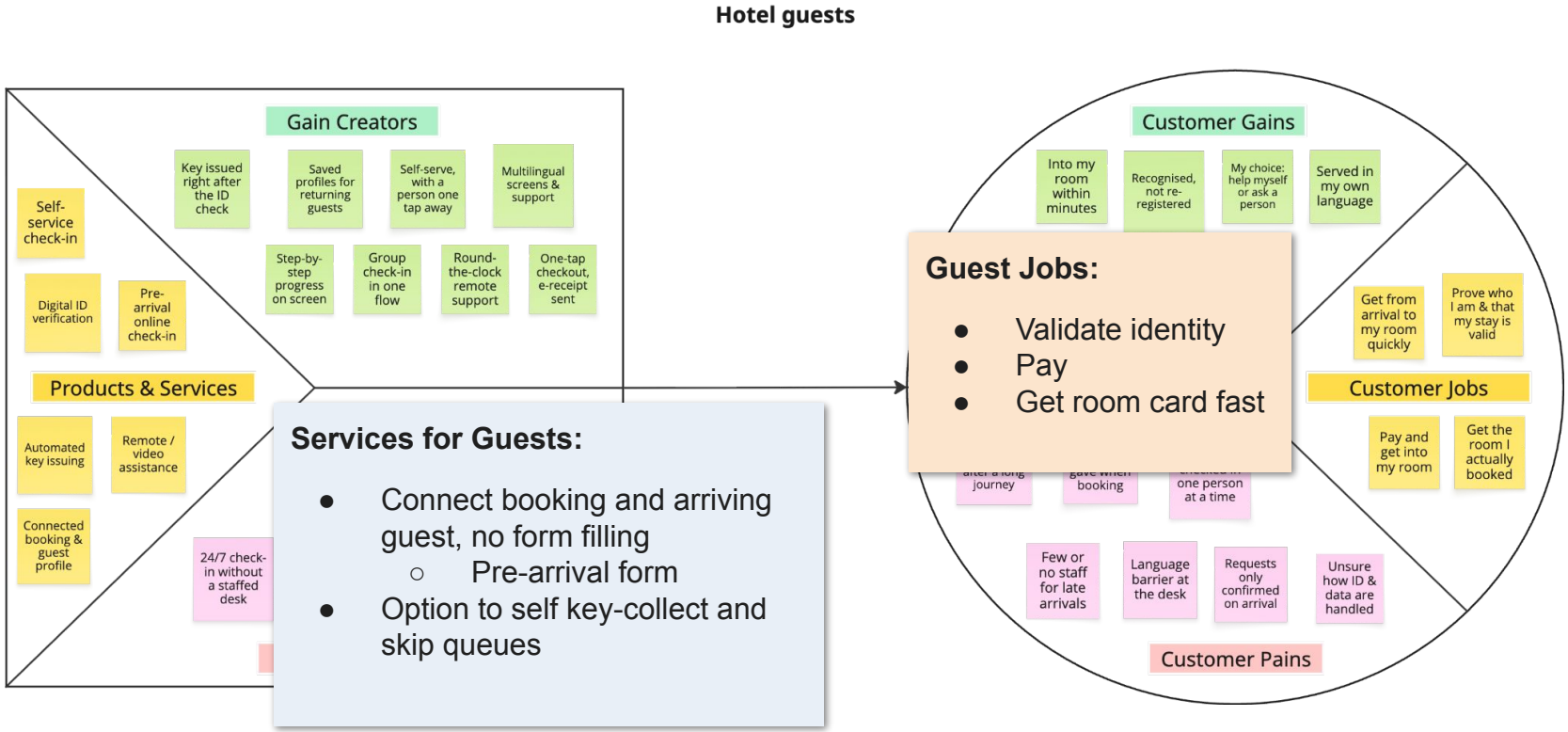
Stakeholders Map



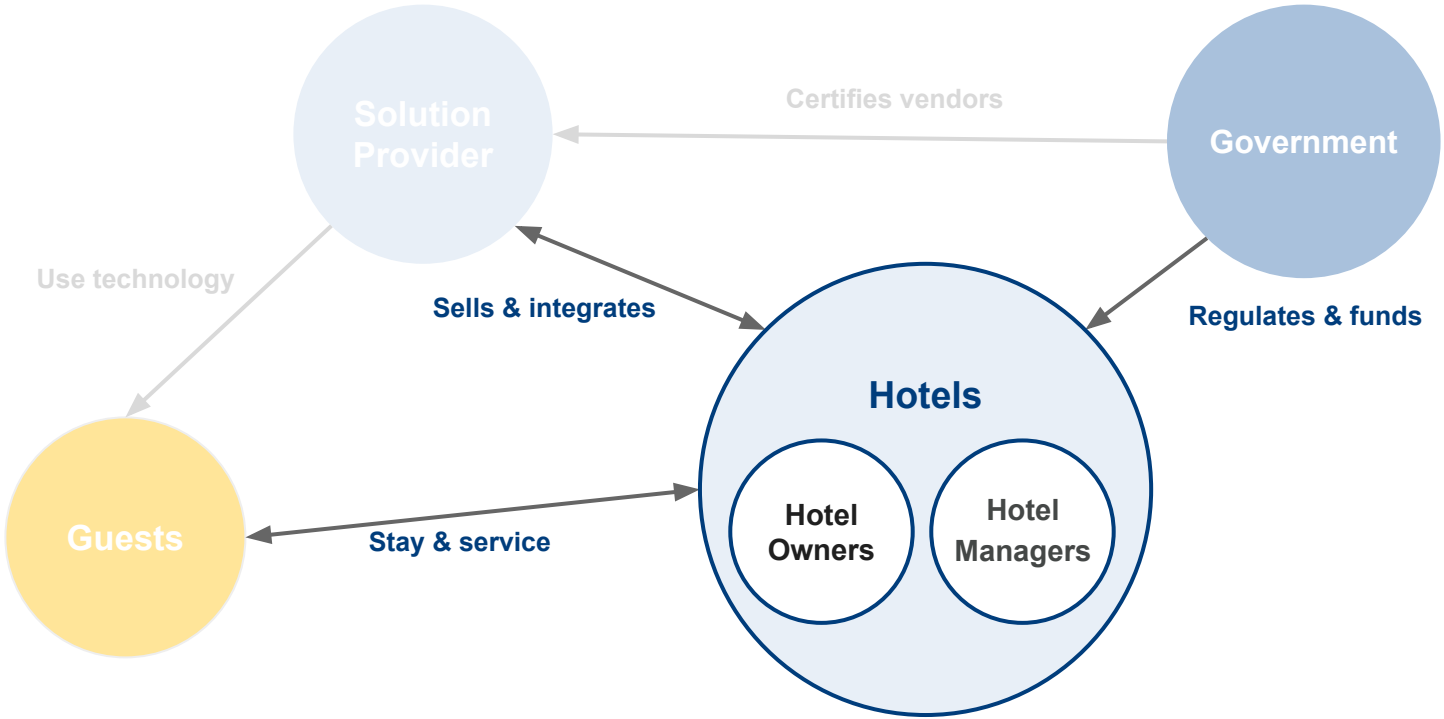
Stakeholders Map



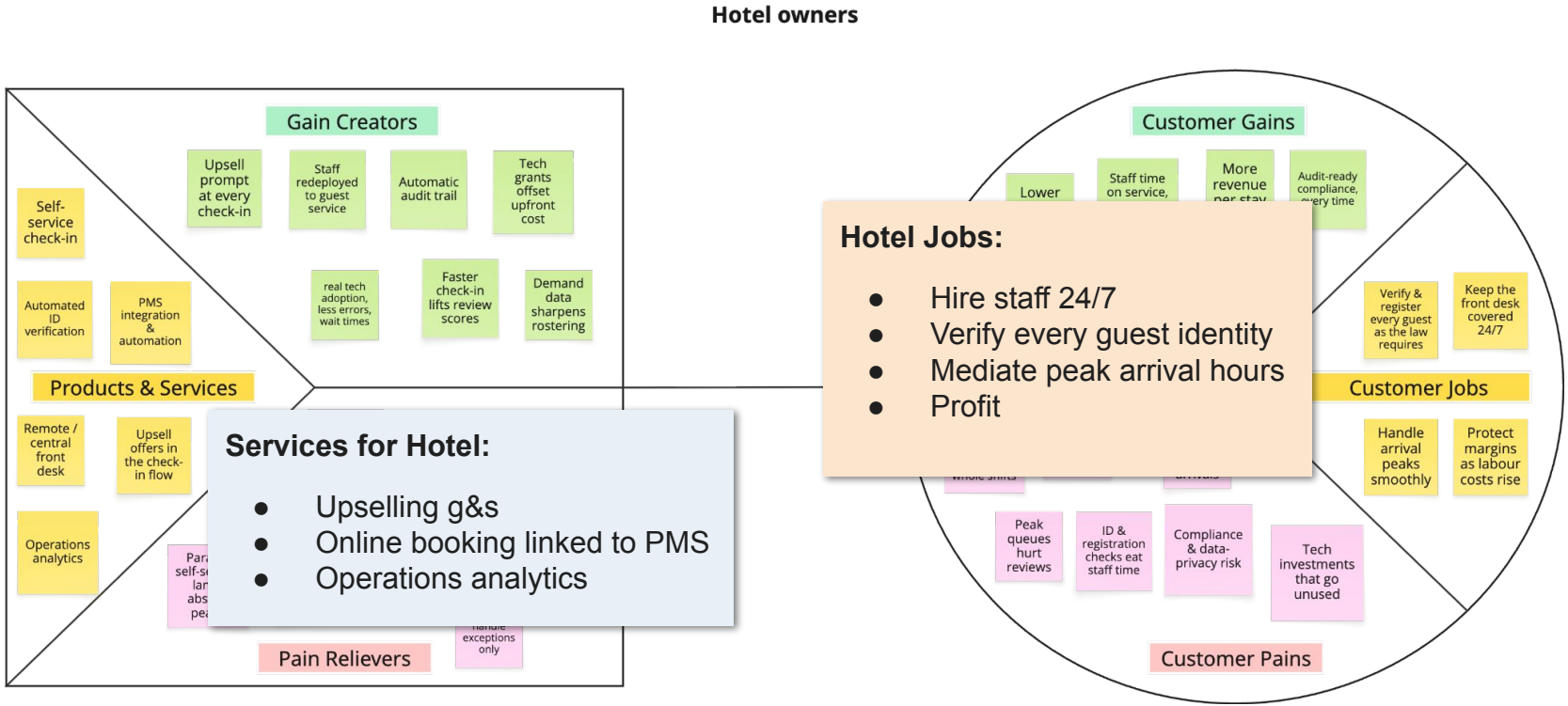
VPC for Guest (Check-in)



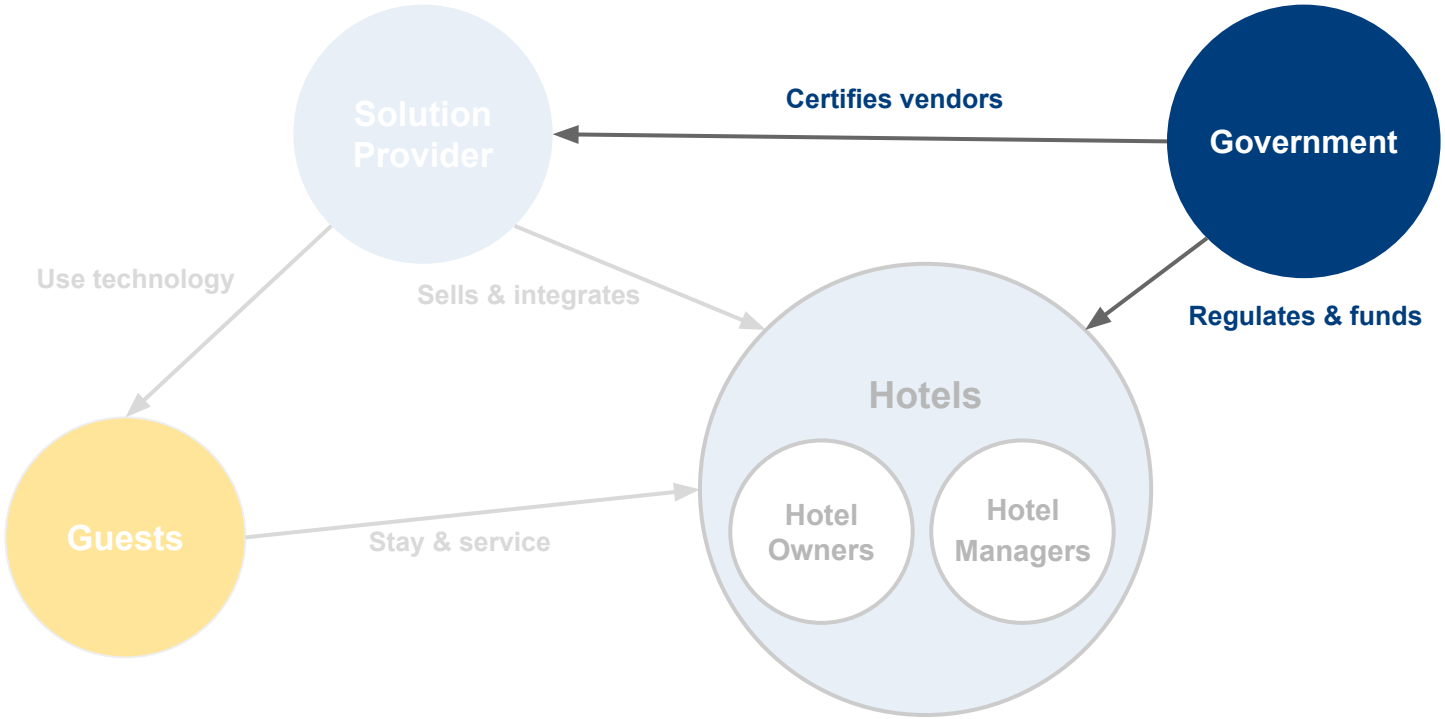
Stakeholders Map



VPC for Hotel (Check-in)



Stakeholders Map



Government & Industry – Push for Digitisation

Government and Industry Push

STB & IMDA Hotel Industry Digital Plan

Smart Rooms	Robotics	**Digital Concierge**	**Digital Check-in/out**	Asset Management System

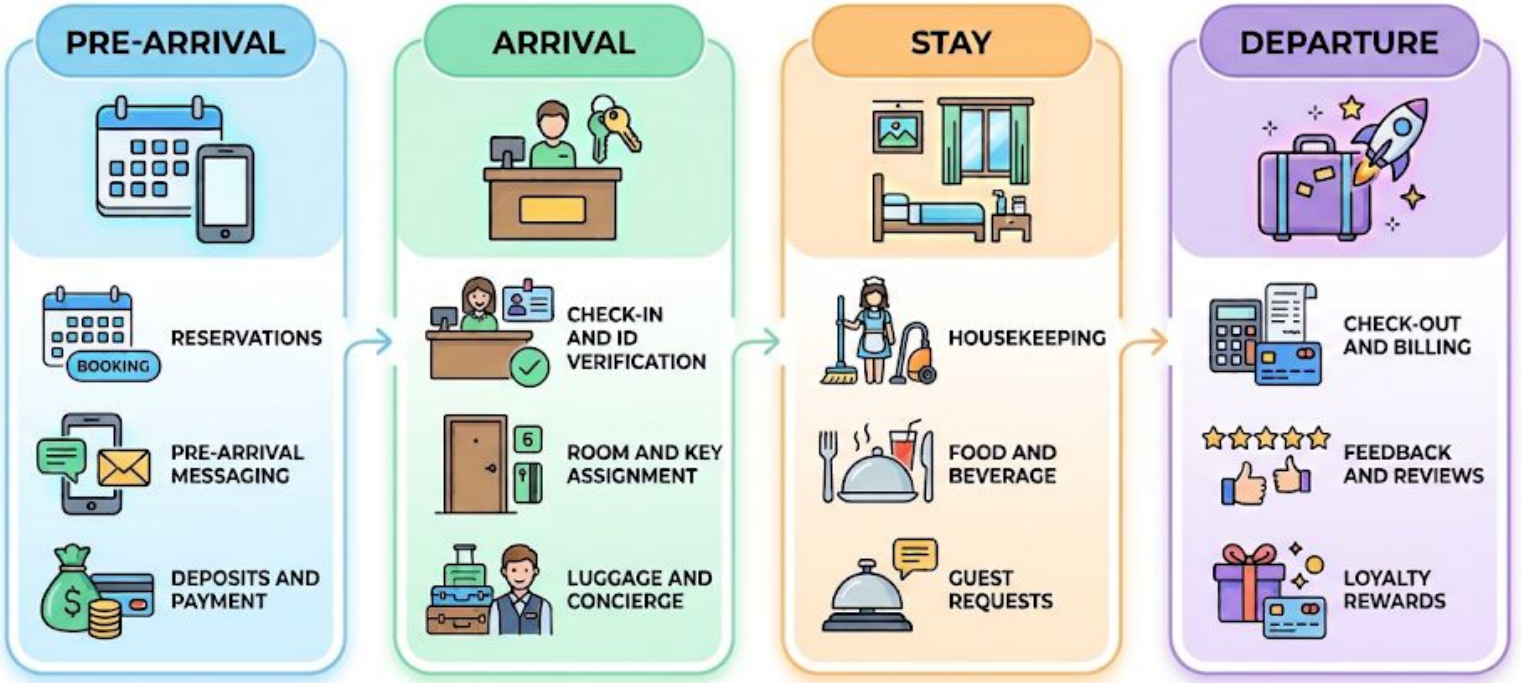
STB Business Improvement Fund

Cash: **70%** funding subsidy (as of 2024)

EVA adoption:

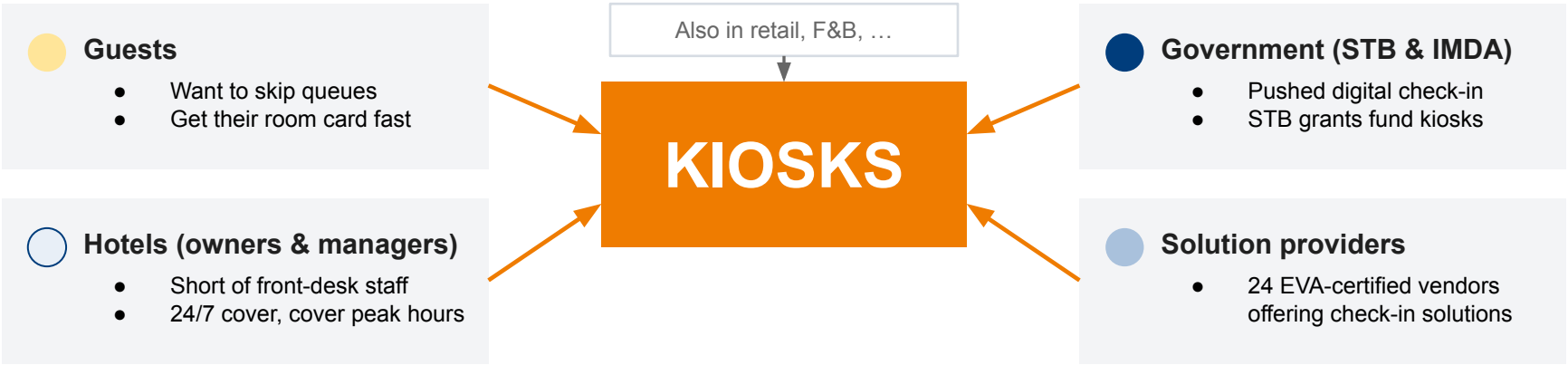
- **24** EVA-certified vendors on STB's list (Sep 2026)
- **70%** Hotel room stock check in/out are connected to EVA

Guest User Journey



The Trend

One recurring pattern we see, not just in hospitality but in other industries facing this manpower shortage, is…



But how's the adoption of kiosks, and what can we learn from it?

Kiosks in other industries



McDonald's F&B

20%

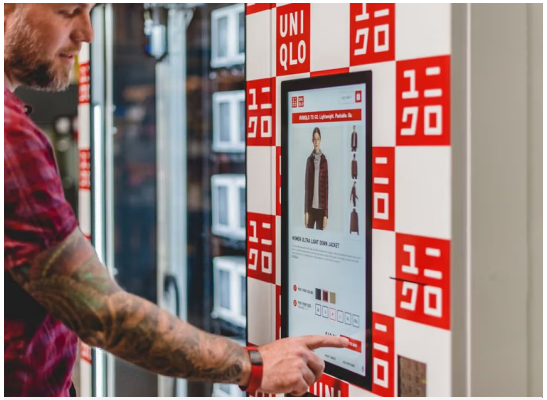
increase in order size



Changi Airport Transport

50%

shorter queue time



Uniqlo Retail

50%

shorter check-outs

Image Sources: McDonald's, Changi Airport, Uniqlo

Problem Statement

How can we reimagine the hotel arrival experience with AI and Emerging Technologies?

Problem Statement

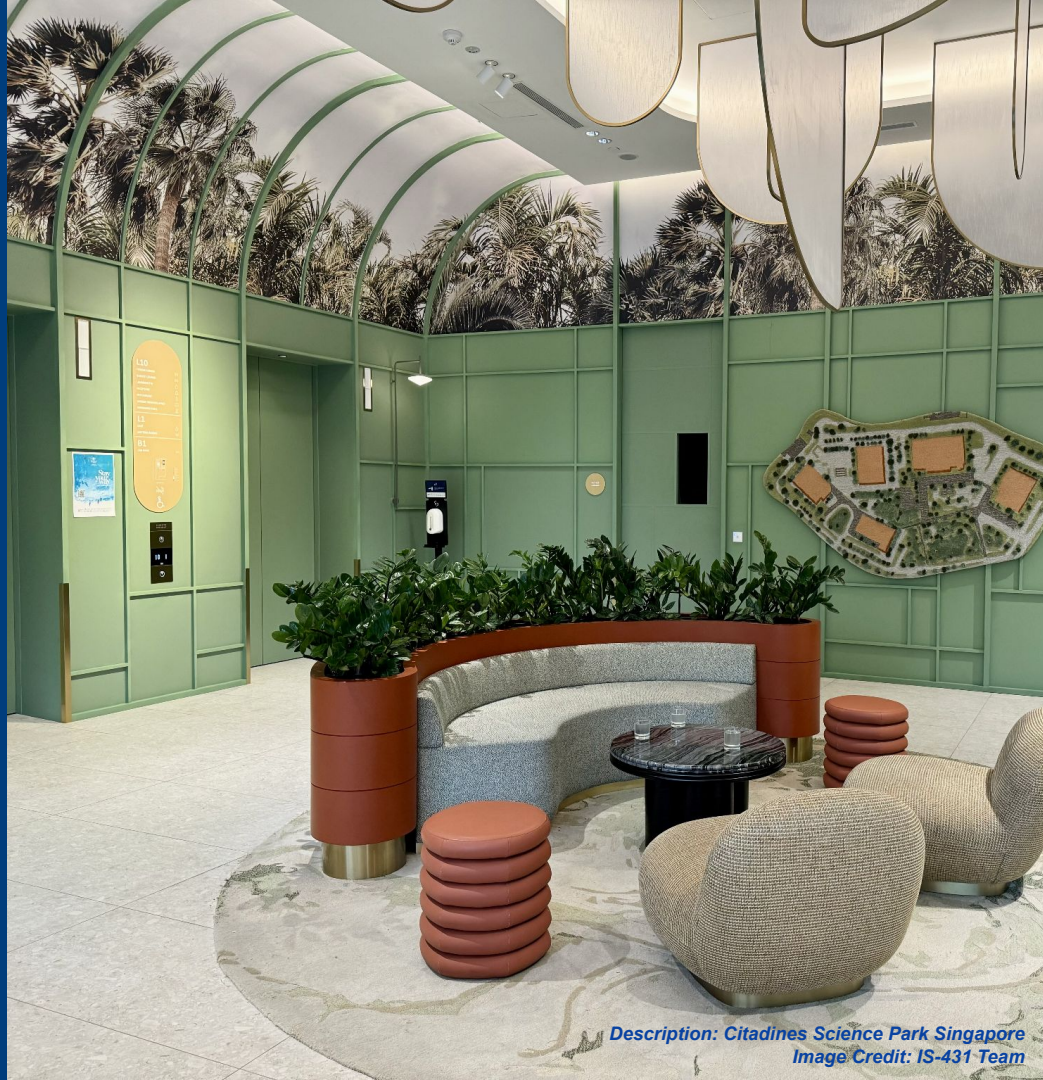
How can we reimagine the hotel arrival experience with AI and Emerging Technologies?

Phase I

How may we improve check-in experience with better kiosk implementation in hotels?

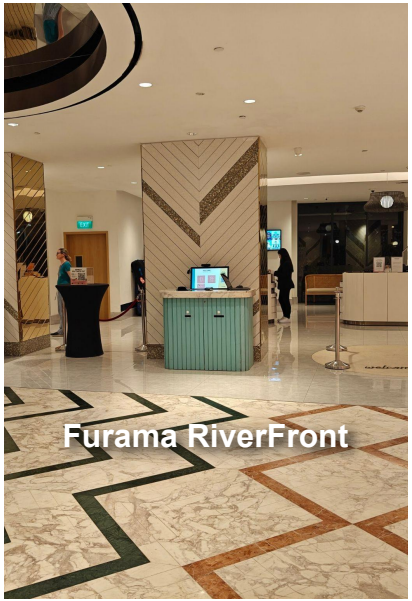
02

Ground Work



Description: Citadines Science Park Singapore
Image Credit: IS-431 Team

Hotel Visits



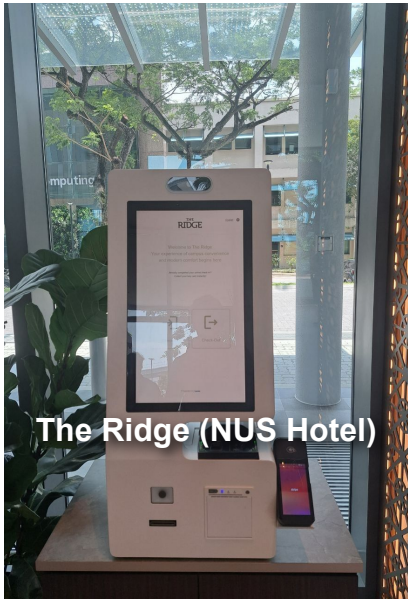
AIT Technologies (AIT)



Jurudata



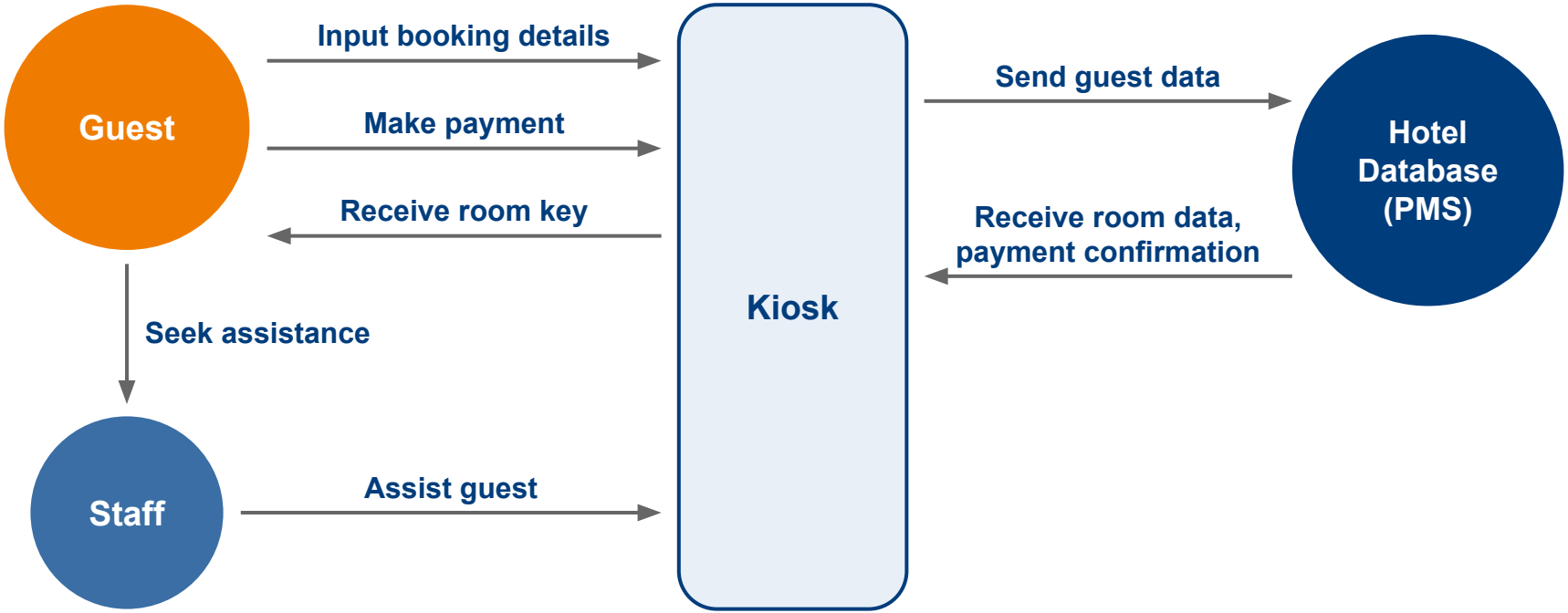
Jurudata



Trevo.ai

Description: Respective Hotel Lobbies
Credit: IS-431 Team

Interaction between Guest and Kiosk



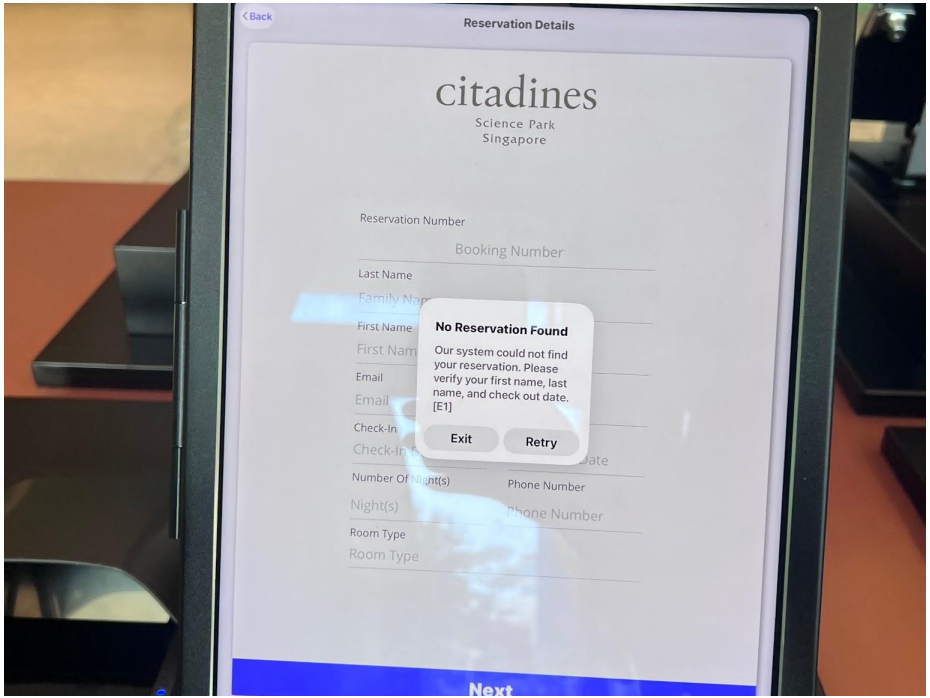
Issues with Kiosks (1 of 3)

1 Basic functionalities

- Passport scan error
- Booking unlinked
- Payment failure
- Key card jammed / missing

2 UI/UX

- Clunky interface
- Multiple screens
- Typing on vertical screen



Citadines Science Park Singapore

Image credit: IS-431 Team

Issues with Kiosks (2 of 3)

3 **Cabling**

- Power plugs
- LAN cables

4 **Position**

- Visible option
- Accessible but private enough



Citadines Rochester Avenue Singapore

Image credit: IS-431 Team

Issues with Kiosks (3 of 3)

5 **Handling large tasks**

- Multi-room bookings
- Many passports

6 **Demand for human touch**

- User preference



Citadines Science Park Singapore

Image credit: IS-431 Team

Hotel Visits - Observations and Identified Challenges

20%

usage rate

Very low usage at certain establishments

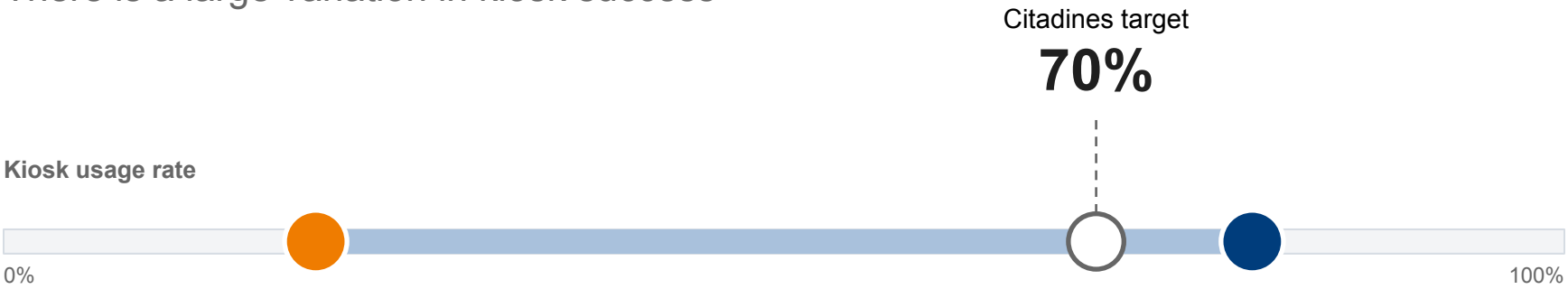
1. Some guests did not notice or use the kiosk at all
2. Some establishments had poor planning and execution of kiosk deployment

⇒ Kiosk design is but one small factor in successful kiosk implementation

Hotel Visits - Observations and Identified Challenges

Not all is doom and gloom.

There is a large variation in kiosk success



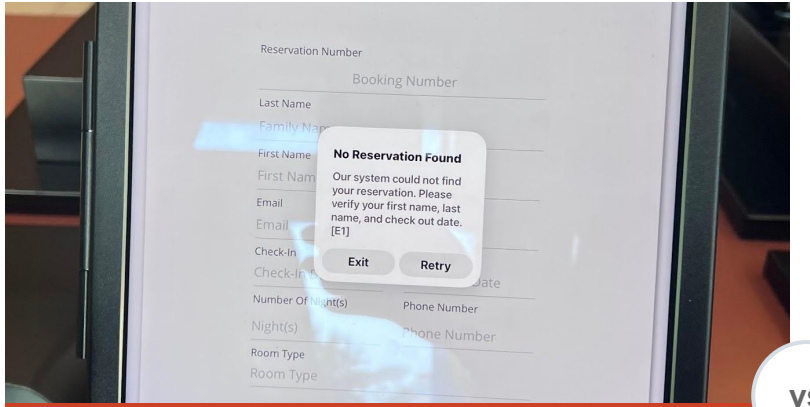
20%

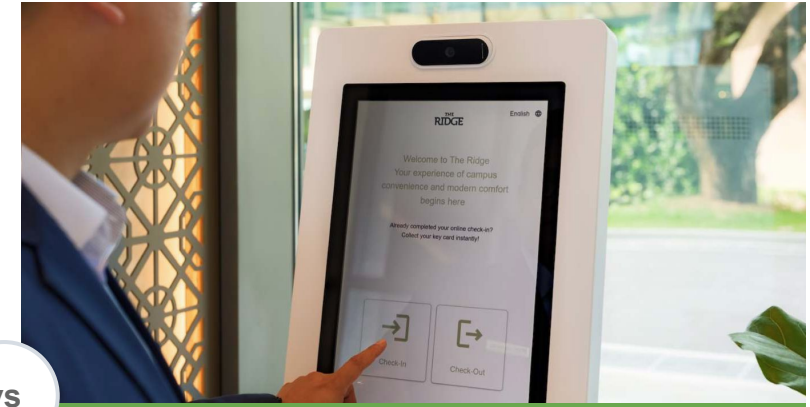
Citadines

85%

The Ridge: acts as a benchmark for a target

Hit or Miss: Kiosk Usage Rates





vs

20%

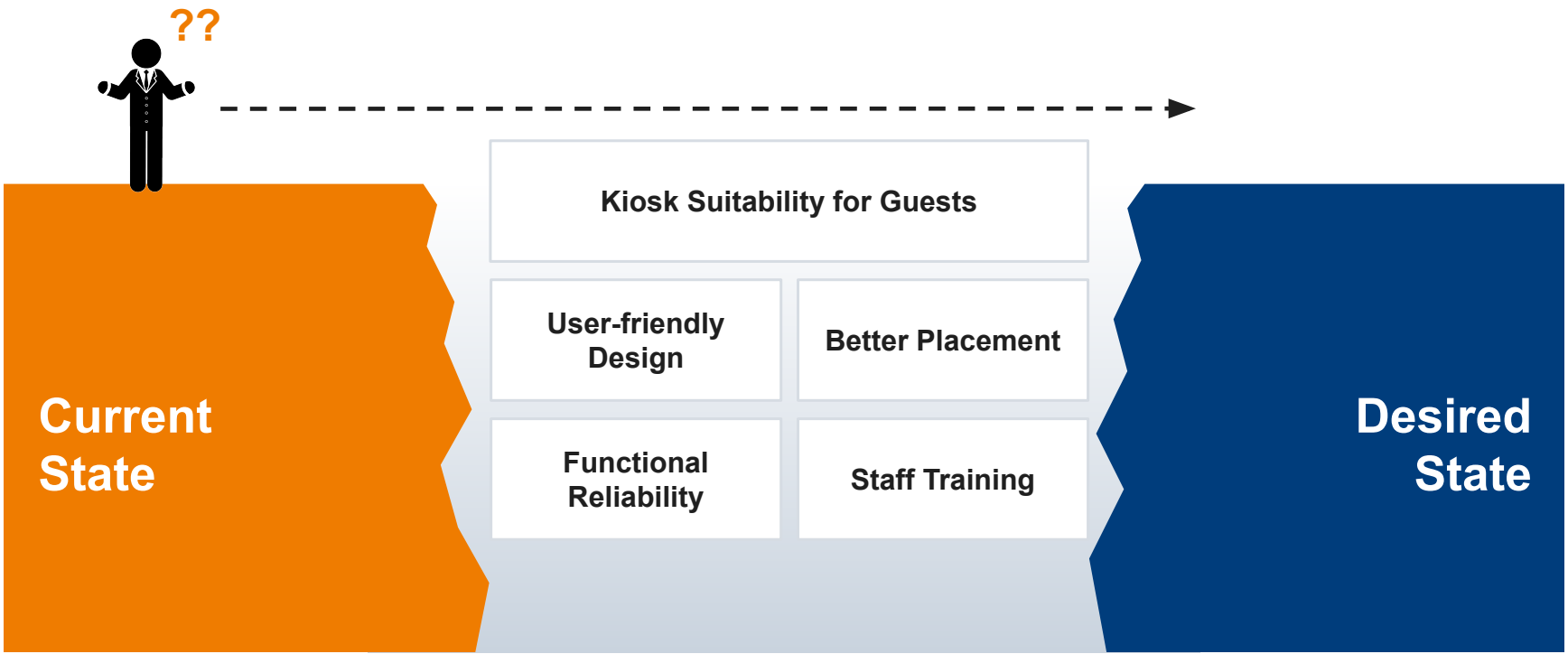
Citadines

85%

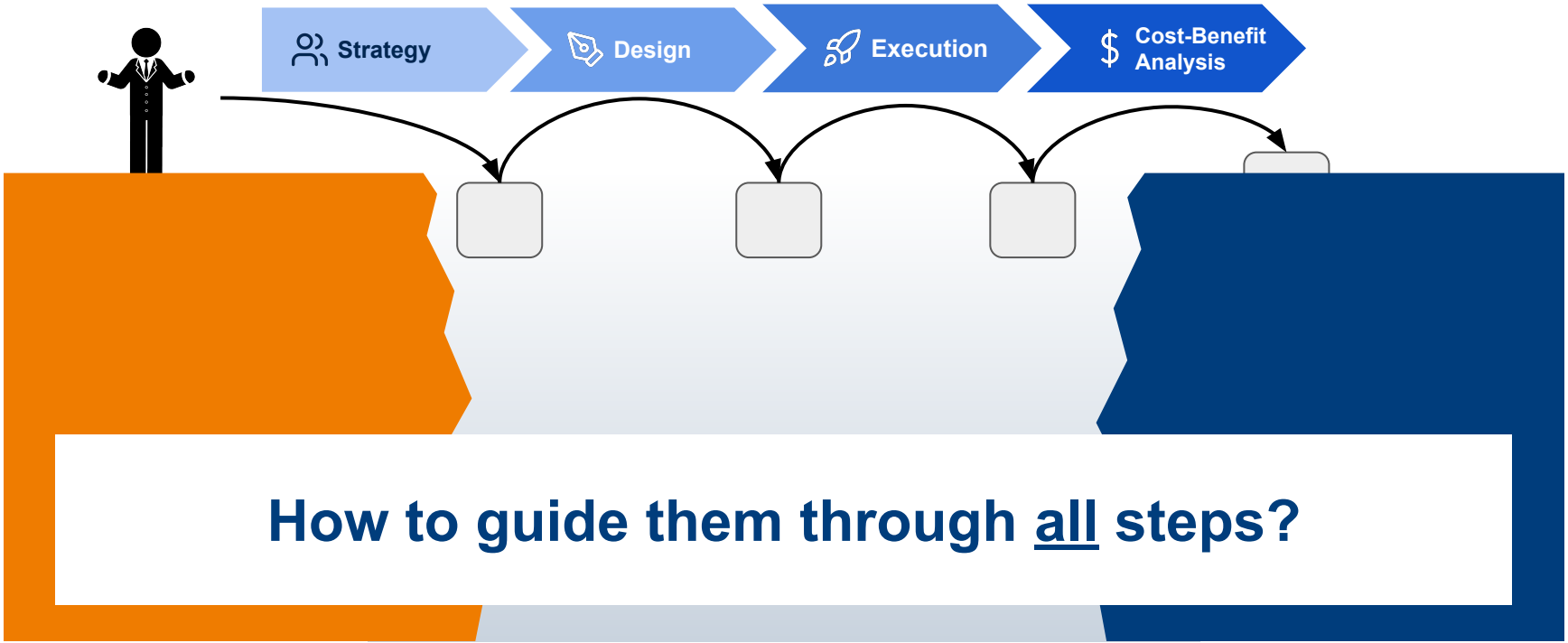
The Ridge

Description: Citadines Science Park Singapore and The Ridge @ NUS
Image Credit: IS-431 Team, The Ridge

Hotel Kiosk Implementation Analysis



Bridging the Gap

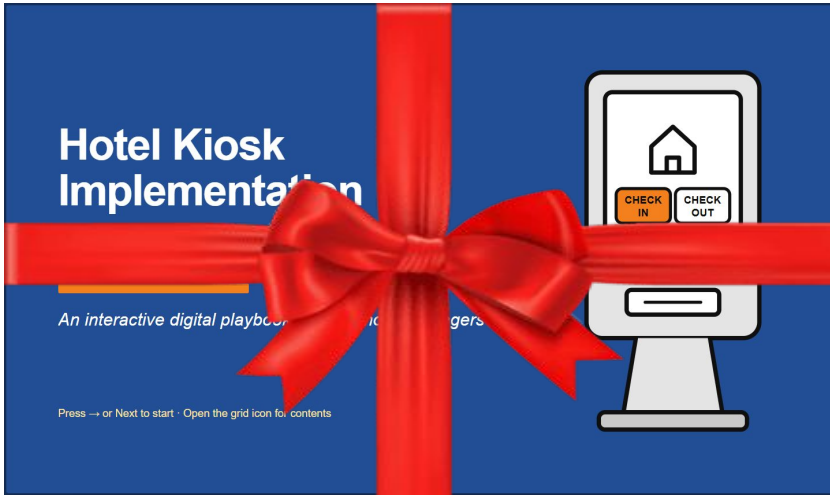


Our Solution – The Playbook

**“Give a man a kiosk, increase check-ins for a day.
Teach a man to kiosk, increase check-ins for life.”**

We made this for you, try it out yourself!





Wonderful!



Playbook Goals

Interactive and Practical





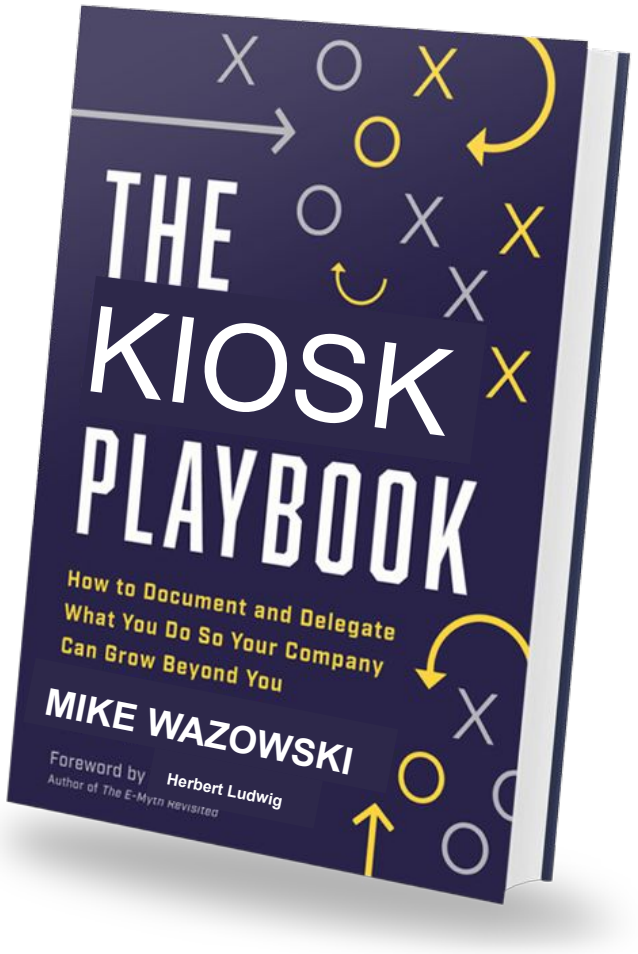
Provide targeted and structured guidance for kiosk implementation

User-centric and adaptable to hotels' specific needs

Helps users go from theory to action

03

Prototype Demonstration



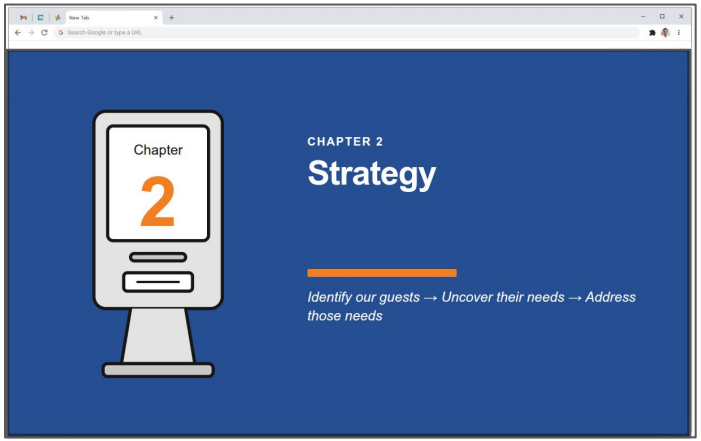
04

The Way Ahead



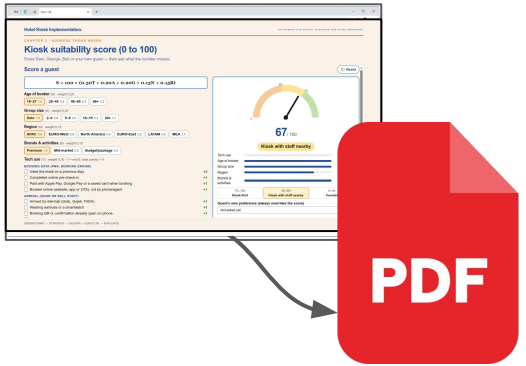
Product Deployment Plans

Platform



Playbook hosted on website

Exporting Data



Completed playbook can be exported to PDF

User Feedback



Mr Sherwin Poh

Senior Manager (Ops)
@ The Ridge NUS

> **“It’s (the playbook’s) great, I would have used it as a guide to make more informed considerations when planning as there may have been things I didn’t think about**

- Kiosk suitability rating can get buy in from superiors
- Aid visualisation of flow and comparison of different placements
- The calculator gives initial financial analysis for STB grant

Way Ahead for the Playbook

Identified Limitations



Lack of scoring data



Limited professional input

Next Steps

- Engage more hotel stakeholders
- Top-down validation from STB and hospitality schools
- Expand scope for **Phase II**

05

Phase II Structure

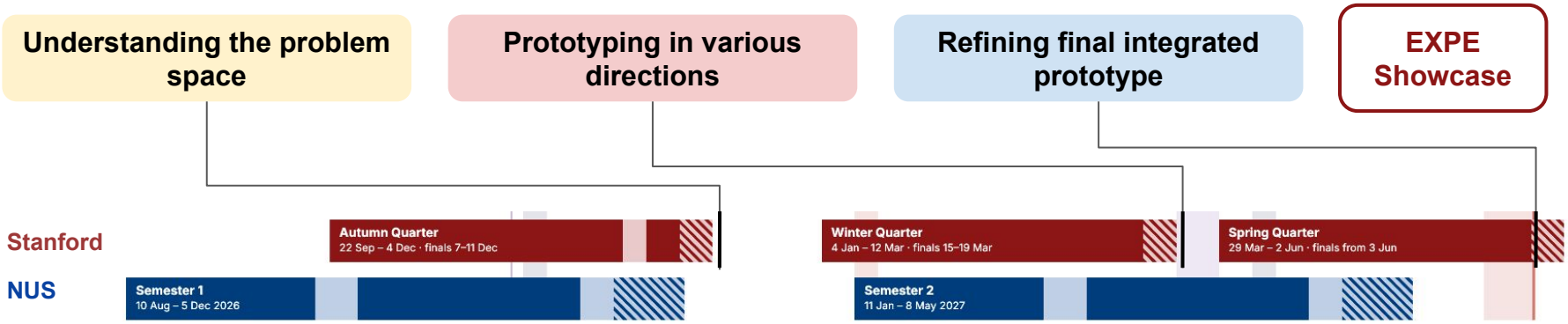


Phase II - Working with ME310

Stanford Global Engineering Design Innovation Program (GEDI)

Stanford ME310 + NUS CDE4301

3 main sprints:



Collaboration Amongst:

Stanford | ENGINEERING
Mechanical Engineering

NUS National University of Singapore
College of Design and Engineering

ASCOTT

VENTURE



×



×



×



Stanford
ME310 Students

NUS
CDE4301 Students

Industry Partner
Problem Owner

Industry Partner
Solution Provider

Phase I's Achievements' Role in Phase II

Phase I

Better kiosk implementation



Build on scope



Reuse implementation principles

Phase II

Reimagine the guest experience

Understanding the problem space

Prototyping in various directions

Refining final integrated prototype

Phase I's Achievements' Role in Phase II

Phase I



Research Foundation
- Industry insights
- Challenges
- Guest journey understanding



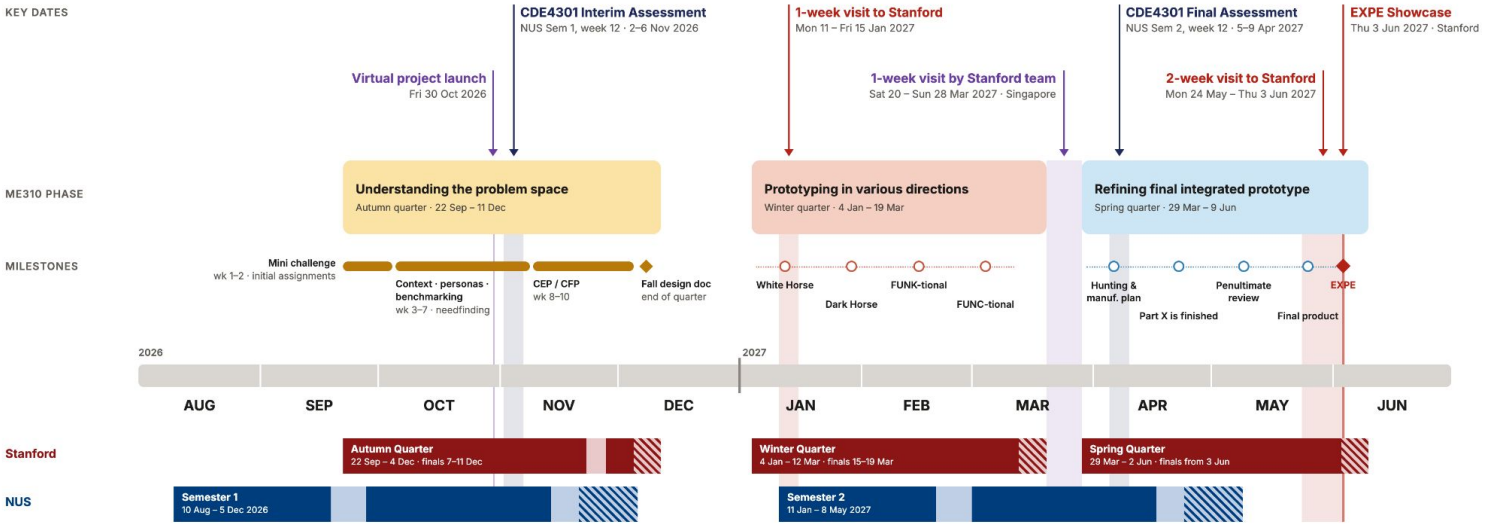
Playbook framework
- Strategy
- Design
- Execution

Phase II

Sprint 1: Understanding the problem space
- Problem identification
- Broader solution ideation

Sprint 3: Refining final integrated prototype
- Principles in playbook can guide prototype design and execution
- Playbook itself can be adapted to become a user manual

ME310 × CDE4301 — AY2026/27 project timeline



Thank you

Any questions?

